Mayank Baranwal

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Education

B. Tech. Computer Science and Engineering (July 2017 --> Present)
Indian Institute of Technolog - Guwahati, India
CGPA: 9.69 (till now)

Courses: Linear Algebra, Discrete Math, Digital Design, Data Structure (Theory + Lab), Real and Complex Analysis, Algorithms*, Automata Theory*, Computer Architecture*, Machine Learning (from Coursera), Hardware Lab*, System SW Lab*, Software Engg Lab* **ongoing courses*

Technical skills

- **Programming**: C, C++, Python, HTML, CSS, Javascript**, MATLAB**
- **Technologies**: mySQL, git, Linux shell utilities, Windows, Django**, Arduino** ***elementary proficiency*

Publications

- **ACISP 2018: 821:830 "CRT-KPS: A Key Pre-distribution Scheme Using CRT", by Pinaki Sarkar, Mayank Baranwal, Sukumar Nandi** | *https://link.springer.com/chapter/10.1007/978-3-319-93638-3_51*
  Applying Chinese Remainder Theorem (CRT) to create non-interactive, highly connected and resilient combinatorial key agreement schemes.

Projects

- **Multi-Lingual Fine-Grained Entity Recognition** (*with Dr. Ashish Anand Assoc. Prof CSE IIT-G*) ***Ongoing***
  Create a neural network capable of classifying an entity mention to a broad set of types though joint learning of representations and label embeddings across multiple languages with transfer-learning strategies.
- **Automatic Guidance System for Motor Vehicles - Texas Instruments IICDC** ***Ongoing***
  Team lead for monitoring surroundings through computer vision to prevent motor vehicle collisions by tracking eye movement and calculating real time minimum braking distance
- **Timetable Generator** | *https://github.com/Mayank-Baranwal/Timetable_Generator* ***Feb 2017 - Mar 2017***
  Semi-Graphical interface to generate timetables automatically under varying constraints for school network.
- **Virtual Enigma Machine** | *https://github.com/Mayank-Baranwal/Enigma_Cryptography* ***Dec 2016***
  Virtual implementation of WW2 German Enigma cipher machine with 5 rotor configurations.

Achievements

- **KRITI 2018-19:** Secured 1st place in the SoilX Operations and Planning Case Study, IIT-Guwahati
- **KVPY 2015-16:** Obtained the National Research Fellowship scholarship, with **All-India-Rank 104**
- **NTSE 2014-15:** Obtained the National Level scholarship
- **KRMO 2014-15** : Qualified for Mathematics 2nd Level Camp at Indian Statistical Institute, Bengaluru
- **US MathCounts' 2013:** Participated at Washington DC, as a part of the four-member global team
- **Helsinki Mathematical Competition' 2012:** Secured 1st place in Helsinki University, at Helsinki, Finland

Extracurricular Activities

- **IAYP Silver Awardee 2016:** 12 months of Community and Service (Selling recyclable bags, teaching)
- **Model United Nations (MUN):** Awarded Best Delegate in the Financial Crisis Committee at KMUN' 2014, Bengaluru, India and Distinguished Delegate at St. Andrew's International MUN' 2013, Dublin, Ireland
- **Tallinn Friendly Olympics 2013:** 1st place in Badminton Championship; 1st place in 4x100 m. relay race